

NILS JANSEN

CURRICULUM VITAE

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AT A GLANCE

Position	Full Professor and Head of the Chair of Artificial Intelligence and Formal Methods, Ruhr University Bochum; part-time Full Professor of Safe and Dependable Artificial Intelligence, Radboud University
Distinctions	ERC Starting Grant · ELLIS Fellow · Principal Investigator, Cluster of Excellence CASA · AAAI Distinguished Paper Award
Research	More than 110 peer-reviewed papers at venues such as NeurIPS, ICLR, IJCAI, AAAI, AAMAS, CAV, and TACAS · more than 4 400 citations · h-index 42
Funding	(Co-)principal investigator of ten funded projects, with a personal budget of more than 3.4 million Euro
Group	Eleven researchers · all three PhD graduates I supervised as main supervisor became postdocs · two former postdocs are assistant professors · ETAPS Doctoral Dissertation Award 2026 for my former PhD student Thom Badings
Leadership	Built the Chair of Artificial Intelligence and Formal Methods at Ruhr University Bochum from scratch (since 2023) · Vice head of department, Radboud University (2023)
Transfer	Projects with Alfa Laval, Canon Production Printing, Ericsson Research, Stateful Robotics, and Multikopter.de · open-source tools COMICS, PROPhESY, and COOL-MC
Service	Associate Editor of JAIR · Area Chair for CAV 2027 · Area Chair or Senior PC at ICLR, AAMAS, IJCAI, and AAAI · QEST Steering Committee (since 2023)

PERSONAL DETAILS AND PROFILES

Nationality	German
Languages	German (native), English (fluent, C2), Dutch (basic, A1)
Google Scholar	scholar.google.com/citations?user=zUavkyEAAAAJ
DBLP	dblp.org/pid/32/8421-1
ORCID	orcid.org/0000-0003-1318-8973

PROFESSIONAL EXPERIENCE

Since Nov 2023	Full Professor, Ruhr University Bochum, Germany Chair of Artificial Intelligence and Formal Methods
Since July 2025	Principal Investigator, Cluster of Excellence CASA, Ruhr University Bochum, Germany CASA: Cyber Security in the Age of Large-Scale Adversaries
Since Sept 2025	Full Professor of Safe and Dependable Artificial Intelligence, part-time, Radboud University, Nijmegen, The Netherlands
July 2022–Aug 2025	Associate Professor, Radboud University, Nijmegen, The Netherlands

Feb 2020–June 2022	Assistant Professor, Radboud University, Nijmegen, The Netherlands Tenured
June 2017–Jan 2020	Assistant Professor, Radboud University, Nijmegen, The Netherlands Tenure track position
Jan 2017–May 2017	Research Associate, University of Texas at Austin, United States Senior researcher in the Autonomous Systems group headed by Prof. Dr. Ufuk Topcu
Jan 2016–Dec 2016	Postdoctoral Research Fellow, University of Texas at Austin, United States Research in the Autonomous Systems group headed by Prof. Dr. Ufuk Topcu
Apr 2015–Jan 2016	Postdoctoral Researcher, RWTH Aachen University, Germany Research and Teaching Assistance at the <i>Software Modeling and Verification Group</i> headed by Prof. Dr. Ir. Dr. h.c. Joost-Pieter Katoen

EDUCATION

March 2015	Dr. rer. nat. in Computer Science, RWTH Aachen, Germany Title of Dissertation: <i>Counterexamples in Probabilistic Verification</i> Final grade: <i>summa cum laude</i> (with distinction) Supervised by Prof. Dr. Erika Ábrahám and Prof. Dr. Ir. Dr. h.c. Joost-Pieter Katoen
March 2009	Diploma in Computer Science, RWTH Aachen, Germany Title of Diploma Thesis: <i>Automaton-definable Tree Relations with Cardinality Constraints</i> Final grade: 1.4 (very good) Supervised by Prof. Dr. Dr. h.c. Wolfgang Thomas

AWARDS AND HONORS

2024	Elected ELLIS Fellow European Laboratory for Learning and Intelligent Systems
2022	ERC Starting Grant DEUCE: Data-Driven Verification and Learning Under Uncertainty
2022	AAAI Distinguished Paper Award Title: <i>Sampling-Based Robust Control of Autonomous Systems with Non-Gaussian Noise</i>
2015	Borchers Badge of RWTH Aachen University for PhD examination with distinction Awarded to the top 10% of the PhD students

THIRD-PARTY FUNDING

Ten projects as (co-)principal investigator, listed with my personal budget. Details on partners and positions are available on request.

Project	Funder and partners	Years	Personal budget
DEUCE : Data-Driven Verification and Learning Under Uncertainty	ERC Starting Grant	2023–2027	1 500 000 Euro
SPARTA : Secure and Privacy-Aware RL Agents	Cluster of Excellence CASA; with V. Moonsamy (RUB)	2026–2029	1 PhD position
FetchFly.AI : Self-learning aerial robotics	ERDF/JTF NRW; Chair of Production Systems (RUB), w3logistics, Multikopter.de	2026–2028	372 900 Euro
PrimaVera : Predictive maintenance	NWA consortium, 4.7 million Euro overall	2019–2025	840 523 Euro
Provably correct policies for uncertain POMDPs	NWO Open Competition (Klein-1)	2020–2024	295 702 Euro
SAM-FMS : Scheduling flexible manufacturing	NWO MasCot; TU/e, TU Delft, Canon Production Printing	2020–2024	223 838 Euro
EXoDuS : Explainable data science	NWA; TNO	2019–2023	100 000 Euro
ORLEANS : Offline RL for sustainable shipping	Radboud IRP; Alfa Laval	2023–2024	50 000 Euro
GRIMES : Green production	Radboud Faculty of Science; Nexperia	2020–2021	26 762 Euro
Safe planning with AI	Radboud–Glasgow Collaboration Fund	2020–2021	12 000 GBP

GROUP MEMBERS: PHD STUDENTS AND POSTDOCTORAL RESEARCHERS

Since 2026	PhD students Florian Kuhm and Sjoerd Jansen Ruhr University Bochum
Since 2025	Postdoctoral Researcher Dr. Maximilian Weininger Ruhr University Bochum
Since 2024	PhD students Miriam Ackermann (née Schäfers), Jule Schmidt, Joshua Wendland, Markel Zubia, Postdoctoral Researcher Dr. Marcel Neuhausen Ruhr University Bochum
Since 2023	PhD students Eline Bovy, Maris Galesloot, and Merlijn Krale Radboud University
2020–2025	PhD students Thom Badings and Marnix Suilen Both graduated in 2025; Thom Badings postdoc at the University of Oxford, now at RWTH Aachen on a fellowship; Marnix Suilen postdoc at the University of Antwerp (FWO fellowship)
2020–2024	PhD student Christoph Schmidl
2021–2023	Postdoctoral Researcher Dr. Thiago Simão Now assistant professor at TU Eindhoven
2019–2023	PhD student Dennis Gross Graduated in 2023; now postdoc at the University of Oslo
2020–2021	Postdoctoral Researcher Dr. Zaharah A. Bukhsh Now assistant professor at TU Eindhoven

AWARDS OF SUPERVISED STUDENTS

2026	ETAPS Doctoral Dissertation Award Thom Badings (PhD 2025, cum laude)
2026	VCLA International Student Award (best Master's thesis) Wietze Koops (Master's thesis, co-supervised with Sebastian Junges)
2026	IPA Dissertation Award Thom Badings
2026	KHMW Kees Schouhamer Immink Prize (Proefschriftprijs) Thom Badings
2025	Honorable Mention, AAAI/ACM SIGAI Doctoral Dissertation Award Thom Badings
2024	KHMW Young Talent Award, Computer Science (first place) Wietze Koops (Master's thesis <i>Algorithms for Dec-POMDPs</i>)
2023	BNAIC/BeNeLearn Best Thesis Award Eline Bovy (Master's thesis)
2023	Christine Mohrmann Stipend for promising female PhD students, Radboud University Eline Bovy
2022	ICAART Best Student Paper Award David Kerkkamp (Master's student), <i>Grouping of Maintenance Actions with Deep Reinforcement Learning and Graph Convolutional Networks</i>

TEACHING

Since 2023	Ruhr University Bochum
First year	Preparatory course in computer science (Python; also organizer) · Introduction to Programming (CS1, Python, from 2027/28) · Programming and Programming Languages · each for several hundred students
Lectures	Programming for Modern Machine Learning (new) · Formal Verification and Model Checking · reinforcement-learning part of Introduction to Artificial Intelligence
Labs, projects	Challenges in Reinforcement Learning (new) · study project on multi-agent reinforcement learning
Seminars	Safety and Reliability in Artificial Intelligence · Theory of Decision-Making under Uncertainty
Since 2017	Radboud University (responsible lecturer and course coordinator)
Bachelor	Processors (mandatory, seven years) · Operating Systems (mandatory, 2017–2020)
Master	Model Checking (since 2017), in the specialization Mathematical Foundations of Computer Science · seminar of that specialization
Supervision	23 Bachelor's and 27 Master's theses (Radboud and Bochum), nine of them published with the student as first author (ICLR, IJCAI, AAAI, ICAPS, ICAART) · 16 research internships

Qualifications

University Teaching Qualification (BKO/UTQ), Radboud University, 2020 · Academic Leadership program for Assistant and Associate Professors, Radboud University, 2023

KEYNOTE TALKS

- 2026 Upcoming keynote (November), 8th International Workshop on Formal Methods for Autonomous Systems (FMAS 2026)
- 2026 Upcoming keynote, IPA Fall Days 2026, Dutch research school IPA (Institute for Programming research and Algorithmics)
- 2025 Keynote, International Conference on Runtime Verification (RV 2025): *Neurosymbolic Learning Systems: Artificial Intelligence and Formal Methods*
- 2025 Keynote, International Conference on Rigorous State-Based Methods (ABZ 2025): *Neurosymbolic Learning Systems: Artificial Intelligence and Formal Methods*
- 2025 Keynote, Research Training Group UnRAVeL, RWTH Aachen University: *Neurosymbolic Learning Systems: Artificial Intelligence and Formal Methods*
- 2023 Keynote, 25th International Symposium on Formal Methods (FM 2023): *Intelligent and Dependable Decision-Making Under Uncertainty*
- 2023 Keynote, 12th Workshop on Synthesis (SYNT 2023), co-located with CAV 2023: *Decision-Making under Uncertainty: Challenges and Highlights*
- 2023 Keynote, Workshop on Verifiable and Robust AI (VRAI), Denmark: *Uncertainty Is Inevitable*

SELECTED INVITED TALKS

- 2026 AI Colloquium, Lamarr Institute and Research Center Trustworthy Data Science and Security, TU Dortmund: *Safe Learning Systems: Artificial Intelligence and Formal Methods*
- 2025 Technion – Israel Institute of Technology: *Neurosymbolic Learning Systems: Artificial Intelligence and Formal Methods*
- 2025 NVIDIA: *Neurosymbolic Intelligent (Reinforcement) Learning Systems*
- 2025 IT University of Copenhagen: *Reliability and Safety in Artificial Intelligence: How Can AI Systems Make Dependable and Safe Decisions Under Uncertainty?*
- 2024 Center for Autonomy Seminar, Oden Institute, UT Austin: *Artificial Intelligence Planning: Uncertainty Is Inevitable*
- 2024 Bosch Center for Artificial Intelligence, Renningen: *Artificial Intelligence Planning: Uncertainty Is Inevitable*
- 2024 Tutorial, Summer School MOVEP 2024: *Decision-Making Under Uncertainty Meets Neurosymbolic AI: Foundations, Challenges, and Highlights*
- 2024 Lecture, Neuro-Symbolic AI Summer School 2024, Centaur AI Institute (online): *A Neurosymbolic Meetup of AI and Formal Methods*
- 2024 Inaugural lecture, Ruhr University Bochum: *Neurosymbolic Intelligent Learning Systems*
- 2023 Colloquium, Institute for Neural Computation, Ruhr University Bochum: *Decision-Making Under Uncertainty*
- 2023 Department of Computer Science, University of Oxford: *Dependable Decision-Making Under Uncertainty: Beyond Probabilities*
- 2023 Digital Futures, KTH Royal Institute of Technology, Stockholm: *Neuro-Symbolic Artificial Intelligence: Intelligent Decisions Under Uncertainty*

2023	2nd Workshop on Open Problems in Learning and Verification of Neural Networks (WOLVERINE), co-located with CAV 2023: <i>Safe Reinforcement Learning</i>
2022	1st Workshop on Safe Reinforcement Learning, co-located with IJCAI 2022: <i>Safe RL: A Collection of Flavors</i>
2022	RL-CONFORM Workshop, co-located with IROS 2022: <i>Safe Reinforcement Learning under Partial Observability</i>
2022	AI Colloquium, TU Dortmund: <i>Intelligent Decision-Making Under Uncertainty: Beyond Probabilities</i>

ORGANIZATION OF SCIENTIFIC EVENTS

2024	Dagstuhl Seminar 24361: Artificial Intelligence and Formal Methods Join Forces for Reliable Autonomy
2023	AISoLA 2023: Track C2 Chair
2023	Dagstuhl Seminar 23492: Model Learning for Improved Trustworthiness in Autonomous Systems
2023	Lorentz Center Workshop: Predictive Maintenance: Let Data Maintain the Model
2022	Lorentz Center Workshop: Rigorous Automated Planning
2021	Flavors of Uncertainty in Verification, Planning, and Optimization (FUNCTION) Co-located with ICALP 2021
2021	National Symposium for Dutch researchers in the field of Software Engineering (SEN) At CWI Amsterdam
2018	Dagstuhl Seminar 18121: Machine Learning and Model Checking Join Forces
2018	Robots, Morality, and Trust through the Verification Lens (ReMOTE) Co-located with CAV 2018
2017	Formal approaches to Explainable VERification (FEVER) Co-located with CAV 2017

EDITORIAL BOARDS

Associate Editor, Journal of Artificial Intelligence Research (JAIR)
 Theme Editor-in-Chief (with Bernhard Steffen) of the theme *Explanation Paradigms Leveraging Analytic Intuition (ExPLAIN)*
 International Journal on Software Tools for Technology Transfer (STTT)

REVIEWING ACTIVITIES

Steering Committee:

Since 2023 QEST (International Conference on Quantitative Evaluation of SysTems)

Program Committees:

2027	CAV (Area Chair)
2026	IJCAI (Senior PC), AAAI (Senior PC), FMCAD, TACAS
2025	ICLR (Area Chair), AAAI, QEST, NeurIPS, NFM
2024	AAMAS (Area Chair), TACAS, AAAI, QEST, NeurIPS

2023	Quantitative Evaluation of Systems (QEST, co-chair), NeurIPS, AAAI, IJCAI, ICLR, ICML, SETTA
2022	AAAI, AAMAS, BNAIC/BeNeLearn, COMPSAC, ICLR, ICML, IJCAI, MOVEP, NeurIPS, QEST
2021	AAAI, AAMAS, ADHS, FASE, ICLR, ICML, FM, NeurIPS, QEST, SPIN
2020	Formal Modeling and Analysis of Timed Systems (FORMATS, co-chair), ICML, IJCAI, NeurIPS, QAVS, SETTA
2019	ICANN, IJCAI, NeurIPS, QEST, SETTA
2018	FORTE, ICE, ICT.OPEN, PhD-iFM, SKILL, TACAS Artifact Evaluation

PhD Reviewer or Committee Member:

Clarissa Costen (University of Oxford), Stefanie Mohr (TU Munich), Pascal van der Vaart (TU Delft), Daniel Scholz (TU Dresden), Jakob Haahr Taankvist (Aalborg University), Mohsen Gaffari (IT University of Copenhagen), Alnis Murtovi (TU Dortmund), Maximilian Schlüter (TU Dortmund), Xiatong Cheng (Ruhr University Bochum), Florent Delgrange (Vrije Universiteit Brussel and University of Antwerp), Théo Matricon (University of Bordeaux), Raphaël Reynouard (Reykjavik University), Kush Grover (TU Munich), Qisong Yang (TU Delft), Zhuofan Xu (ENS Paris-Saclay), Steven Carr (UT Austin), Simon Jantsch (TU Dresden), Ivan Gavran (MPI-SWS), Leonore Winterer (University of Freiburg), Alexis Linard (Radboud University), Rick Smetsers (Radboud University), Konstantin Kueffner (IST Austria)

Habilitation Committee Member:

Petr Novotný and Vít Musil (Masaryk University, Brno)

Reviewer of Grant Applications:

ERC Consolidator Grants, French National Research Agency (ANR), Vienna Science and Technology Fund (WWTF), German Research Foundation (DFG), Novo Nordisk Foundation, Schmidt Sciences

Journal Reviewer:

ACM Transactions on Modeling and Computer Simulation, ACM Transactions on Computational Logic, Acta Informatica, Journal of Artificial Intelligence Research, International Journal of Robotics Research, Science of Computer Programming, IEEE Control Systems Letters, IEEE Transactions on Reliability, IEEE Transactions on Automatic Control, Journal of Automated Reasoning, Journal of Systems and Software, Theoretical Computer Science

External reviewer:

ACC, ATVA, CAV, CDC, CONCUR, FACS, FM, FMOODS–FORTE, FORMATS, FOSSACS, FSEN, FSTTCS, FTSCS, HSCC, IFM, Petri Nets, QEST, SIMULTECH, TACAS, TASE, VMCAI

INSTITUTIONAL RESPONSIBILITIES

Current	Coordinator of school outreach, Faculty of Computer Science, Ruhr University Bochum
Current	Member of the advisory council (Beraterkreis) of the Faculty of Computer Science, Ruhr University Bochum
Since 2025	Member of the Academic Advisory Board (Studienbeirat) of the Faculty of Computer Science, Ruhr University Bochum
2026	Appointment committee for a professorship in Quantum Computing at Ruhr University Bochum
2026	Appointment committee for a lecturer in Practical Computer Science at Ruhr University Bochum

2023	Vice head of the Department of Software Science at Radboud University Nijmegen
2022	Appointment committee for a full professorship in Data Science at Radboud University
2022	Chair of the appointment committee for a professorship in Machine Learning at Radboud University
2020	Appointment committee for several professorships in Digital Security at Radboud University
2019–2023	Organization of the internal seminars at the Software Science Institute at Radboud University
2018–2023	Member of the Program Committee for teaching (Opleidingscommissie) at Radboud University
2014–2016	Member of the examination board in Computer Science at RWTH Aachen University
2014–2016	Member of the commission for teaching in Computer Science at RWTH Aachen University
2010–2011	Steering committee member of the graduate school AlgoSyn at RWTH Aachen University

MEMBERSHIP IN PROFESSIONAL ORGANIZATIONS

Since 2024	ELLIS Fellow European Laboratory for Learning and Intelligent Systems
Since 2020	ELLIS Member European Laboratory for Learning and Intelligent Systems
Since 2019	VERSEN Dutch National Association for Software Engineering
Since 2016	ETAPS European Joint Conferences on Theory and Practice of Software

MEDIA

2024	MIT Technology Review: Mathematik KI braucht dringend Nachhilfe authored by Eike Kühl
2022	AIhub: Developing safe controllers for autonomous systems under uncertainty authored by Thom Badings and Nils Jansen, published by the Association for the Understanding of Artificial Intelligence
2021	Communications of the ACM: Approach to AI Offers More Certainty in the Face of Uncertainty
2023	I/O magazine: How can AI better handle uncertainties in the real world? authored by Bennie Mols
2023	TechGelderland: Using data to make the world safer authored by Joep Peters
2021	Innovation Origins: A new model reduces uncertainty in AI authored by Cindy Cloin
2021	Computable: Radboud maakt kunstmatige intelligentie slimmer Alfred Monterie

2021	Bits&Chips: Radboud and TUE help improve AI decision-making Nieke Roos
2021	Executive-People: Nieuwe AI benadering biedt meer zekerheid bij onzekerheid

PUBLICATIONS

So far, I have published more than 110 peer-reviewed articles in conference proceedings and journals. According to Google Scholar (September 2026), I have more than 4 400 citations (h-index 42, i10-index 80).

BOOK CHAPTERS, EDITED VOLUMES, AND PHD THESIS

- [1] **Nils Jansen**, Sebastian Junges, Benjamin Lucien Kaminski, Christoph Matheja, Thomas Noll, Tim Quatmann, Mariëlle Stoelinga, Matthias Volk (Eds.): Principles of Verification: Cycling the Probabilistic Landscape - Essays Dedicated to Joost-Pieter Katoen on the Occasion of His 60th Birthday, Part I. Lecture Notes in Computer Science 15260, Springer 2024.
- [2] **Nils Jansen**, Sebastian Junges, Benjamin Lucien Kaminski, Christoph Matheja, Thomas Noll, Tim Quatmann, Mariëlle Stoelinga, Matthias Volk (Eds.): Principles of Verification: Cycling the Probabilistic Landscape - Essays Dedicated to Joost-Pieter Katoen on the Occasion of His 60th Birthday, Part II. Lecture Notes in Computer Science 15261, Springer 2024.
- [3] **Nils Jansen**, Sebastian Junges, Benjamin Lucien Kaminski, Christoph Matheja, Thomas Noll, Tim Quatmann, Mariëlle Stoelinga, Matthias Volk (Eds.): Principles of Verification: Cycling the Probabilistic Landscape - Essays Dedicated to Joost-Pieter Katoen on the Occasion of His 60th Birthday, Part III. Lecture Notes in Computer Science 15262, Springer 2024.
- [4] Murat Cubuktepe, **Nils Jansen**, Ufuk Topcu: Shared Control with Human Trust and Workload Models. **Cyber-Physical-Human Systems** 2023.
- [5] **Nils Jansen**, Mirco Tribastone (Eds.): Quantitative Evaluation of Systems - 20th International Conference, QEST 2023, Antwerp, Belgium, September 20-22, 2023, Proceedings. Lecture Notes in Computer Science 14287, Springer 2023.
- [6] **Nils Jansen**, Mariëlle Stoelinga, Petra van den Bos (Eds.): A Journey from Process Algebra via Timed Automata to Model Learning - Essays Dedicated to Frits Vaandrager on the Occasion of His 60th Birthday. Lecture Notes in Computer Science 13560, Springer 2022.
- [7] Nathalie Bertrand, **Nils Jansen** (Eds.): Formal Modeling and Analysis of Timed Systems - 18th International Conference, FORMATS 2020, Vienna, Austria, September 1-3, 2020, Proceedings. Lecture Notes in Computer Science 12288, Springer 2020.
- [8] Radu Calinescu, Marco Autili, Javier Cámara, Antinisca Di Marco, Simos Gerasimou, Paola Inverardi, Alexander Perucci, **Nils Jansen**, Joost-Pieter Katoen, Marta Z. Kwiatkowska, Ole J. Mengshoel, Romina Spalazzese, Massimo Tivoli: Synthesis and Verification of Self-aware Computing Systems. **Self-Aware Computing Systems** 2017: 337–373.
- [9] **Nils Jansen**: Counterexamples in probabilistic verification. **PhD thesis**, RWTH Aachen University, Germany, 2015.

JOURNAL PUBLICATIONS

- [10] Zaharah Allah Bukhsh, Hajo Molegraaf, **Nils Jansen**: A maintenance planning framework using online and offline deep reinforcement learning. **Neural Comput. Appl.** 37(19): 13209–13220 (2025).
- [11] Bettina Könighofer, Roderick Bloem, **Nils Jansen**, Sebastian Junges, Stefan Pranger: Shields for Safe Reinforcement Learning. **Commun. ACM** 68(11): 80–90 (2025).
- [12] Sebastian Junges, Erika Ábrahám, Christian Hensel, **Nils Jansen**, Joost-Pieter Katoen, Tim Quatmann, Matthias Volk: Parameter synthesis for Markov models: covering the parameter space. **Formal Methods Syst. Des.** 62(1): 181–259 (2024).
- [13] Leonore Winterer, Ralf Wimmer, Bernd Becker, **Nils Jansen**: Strong Simple Policies for POMDPs. **Int. J. Softw. Tools Technol. Transf.** 26(3): 269–299 (2024).

- [14] Thom Badings, Licio Romao, Alessandro Abate, David Parker, Hasan A. Poonawala, Mariëlle Stoelinga, **Nils Jansen**: Robust Control for Dynamical Systems with Non-Gaussian Noise via Formal Abstractions. **J. Artif. Intell. Res.** 76: 341–391 (2023).
- [15] Thom Badings, Thiago D. Simão, Marnix Suilen, **Nils Jansen**: Decision-making under uncertainty: beyond probabilities. **Int. J. Softw. Tools Technol. Transf.** 25(3): 375–391 (2023).
- [16] **Nils Jansen**, Gerrit Nolte, Bernhard Steffen: Explanation Paradigms Leveraging Analytic Intuition (ExPLAIIn). **Int. J. Softw. Tools Technol. Transf.** 25(3): 241–247 (2023).
- [17] Thom Badings, Murat Cubuktepe, **Nils Jansen**, Sebastian Junges, Joost-Pieter Katoen, Ufuk Topcu: Scenario-based verification of uncertain parametric MDPs. **Int. J. Softw. Tools Technol. Transf.** 24(5): 803–819 (2022).
- [18] Murat Cubuktepe, **Nils Jansen**, Sebastian Junges, Joost-Pieter Katoen, Ufuk Topcu: Convex Optimization for Parameter Synthesis in MDPs. **IEEE Trans. Autom. Control.** 67(12): 6333–6348 (2022).
- [19] Mohamadreza Ahmadi, **Nils Jansen**, Bo Wu, Ufuk Topcu: Control Theory Meets POMDPs: A Hybrid Systems Approach. **IEEE Trans. Autom. Control.** 66(11): 5191–5204 (2021).
- [20] Zaharah Allah Bukhsh, **Nils Jansen**, Aaqib Saeed: Damage detection using in-domain and cross-domain transfer learning. **Neural Comput. Appl.** 33(24): 16921–16936 (2021).
- [21] Steven Carr, **Nils Jansen**, Ufuk Topcu: Task-Aware Verifiable RNN-Based Policies for Partially Observable Markov Decision Processes. **J. Artif. Intell. Res.** 72: 819–847 (2021).
- [22] Murat Cubuktepe, **Nils Jansen**, Mohammed Alshiekh, Ufuk Topcu: Synthesis of Provably Correct Autonomy Protocols for Shared Control. **IEEE Trans. Autom. Control.** 66(7): 3251–3258 (2021).
- [23] Hadas Kress-Gazit, Kerstin Eder, Guy Hoffman, Henny Admoni, Brenna Argall, Rüdiger Ehlers, Christoffer Heckman, **Nils Jansen**, Ross A. Knepper, Jan Křetínský, Shelly Levy-Tzedek, Jamy Li, Todd D. Murphey, Laurel D. Riek, Dorsa Sadigh: Formalizing and guaranteeing human-robot interaction. **Commun. ACM** 64(9): 78–84 (2021).
- [24] Leonore Winterer, Sebastian Junges, Ralf Wimmer, **Nils Jansen**, Ufuk Topcu, Joost-Pieter Katoen, Bernd Becker: Strategy Synthesis for POMDPs in Robot Planning via Game-Based Abstractions. **IEEE Trans. Autom. Control.** 66(3): 1040–1054 (2021).
- [25] Bram Ton, Rob Basten, John Bolte, Jan Braaksma, Alessandro Di Bucchianico, Philippe Calseyde, Frank Grooteman, Tom Heskes, **Nils Jansen**, Wouter Teeuw, Tiedo Tinga, Mariëlle Stoelinga: PrimaVera: Synergising Predictive Maintenance. **Applied Sciences** 10 (2020).
- [26] Federico Olmedo, Friedrich Gretz, **Nils Jansen**, Benjamin Lucien Kaminski, Joost-Pieter Katoen, Annabelle McIver: Conditioning in Probabilistic Programming. **ACM Trans. Program. Lang. Syst.** 40(1): 4:1–4:50 (2018).
- [27] Ralf Wimmer, **Nils Jansen**, Andreas Vorpahl, Erika Ábrahám, Joost-Pieter Katoen, Bernd Becker: High-level Counterexamples for Probabilistic Automata. **Log. Methods Comput. Sci.** 11(1) (2015).
- [28] **Nils Jansen**, Ralf Wimmer, Erika Ábrahám, Barna Zajzon, Joost-Pieter Katoen, Bernd Becker, Johann Schuster: Symbolic counterexample generation for large discrete-time Markov chains. **Sci. Comput. Program.** 91: 90–114 (2014).
- [29] Ralf Wimmer, **Nils Jansen**, Erika Ábrahám, Joost-Pieter Katoen, Bernd Becker: Minimal counterexamples for linear-time probabilistic verification. **Theor. Comput. Sci.** 549: 61–100 (2014).

CONFERENCE AND WORKSHOP PROCEEDINGS

- [30] Asger Horn Brorholt, Maris F. L. Galesloot, **Nils Jansen**, Kim Guldstrand Larsen, Christian Schilling: Adaptive Probabilistic Shielding by Learning MDPs for Safe Reinforcement Learning. **RV** 2026 (to appear).
- [31] Kush Grover, Markel Zubia, Debraj Chakraborty, Muqsit Azeem, **Nils Jansen**, Jan Křetínský: Resilient Strategies for Stochastic Systems: How Much Does it Take to Break a Winning Strategy? **AAMAS** 2026: 3814–3822.
- [32] David Hudák, Maris F. L. Galesloot, Martin Tappler, Martin Kuřečka, **Nils Jansen**, Milan Česka: Finite-State Controllers for (Hidden-Model) POMDPs using Deep Reinforcement Learning. **AAMAS** 2026: 1201–1210.
- [33] Miriam Schäfers, Merlijn Krale, Thiago D. Simão, **Nils Jansen**, Maximilian Weininger: Perception-Based Beliefs for POMDPs with Visual Observations. **AAMAS** 2026: 3051–3059.

- [34] Jule Schmidt, Xin Tao, Chelsea Sidrane, Swarup Kumar Mohalik, Akhil Prasad, Jana Tumova, **Nils Jansen**: A Case for Causal Reinforcement Learning in Longitudinal Vehicle Control. **ICAART** 2026: 2759–2766.
- [35] Jule Schmidt, Maximilian Weininger, Clemens Dubslaff, David Parker, **Nils Jansen**: Property-driven Causal Abstractions for Markov Decision Processes. **FMCAD** 2026 (to appear).
- [36] Joshua Wendland, Markel Zubia, Roman Andriushchenko, Maris F. L. Galesloot, Milan Češka, Henrik von Kleist, Thiago D. Simão, Maximilian Weininger, **Nils Jansen**: Missingness-MDPs: Bridging the Theory of Missing Data and POMDPs. **IJCAI** 2026: 6217–6226.
- [37] Erika Ábrahám, **Nils Jansen**, Simon Jantsch: Probabilistic Counterexamples Through the Ages. **Principles of Formal Quantitative Analysis** 2025: 174–196.
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